

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL
INTELLIGENCE
AND EXPERT SYSTEMS

COURSE CODE: MLA01
04

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I S H A M S . (1 9 2 3 2 5 0 7 6) ____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

SHAM S.

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialog flow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Problem Analysis

Universities receive a large number of students queries every day related to course registration, examination schedules, class timetables, attendance records, and campus services. Traditionally, these queries are handled by faculty members or administrative staff. This manual process is time-consuming, increases workload, and may delay responses during busy academic periods.

Artificial Intelligence provides an effective solution by developing an intelligent chatbot that can understand student questions and provide instant responses. A Dialog flow-based Student Support Assistant can communicate with students 24×7, reducing human effort while improving the overall quality of student support.

Agent Design

Objective

- To provide instant, accurate, and automated responses to student queries.

Users

- Students
- Faculty Members
- Administrative Staff

Environment

- University Website
- Mobile Application
- Computer Systems

Inputs

- Student text or voice queries

Outputs

- Correct information regarding registration, timetable, attendance, exams, and campus services.

Performance Measures

- High response accuracy
- Fast response time
- User satisfaction
- Easy interaction

Dialog flow Agent Development

Dialog flow is Google's Natural Language Processing (NLP) platform used to develop conversational AI applications. It identifies the user's intent and generates suitable responses.

Intents

- Welcome Intent
- Course Registration
- Timetable
- Attendance Details
- Examination Schedule
- Campus Services
- Default Fallback Intent

Entities

Entities are used to extract important information from user queries.

Examples include:

- Course Name
- Department
- Semester
- Student ID

Training Phrases

- "Show my timetable."
- "When are my exams?"
- "Check my attendance."
- "Register for Artificial Intelligence."
- "Where is the library?"

Responses

Each intent contains predefined responses that are displayed when the user's query matches the intent.

Working Process

Student Query

↓

Intent Detection

↓

Entity Recognition

↓

Context Processing

↓

Information Retrieval

↓

Generate Response

↓

Display Answer

(Insert screenshots of your Dialog flow agent here.)

Problem Formulation and Conversation Flow

The chatbot interaction can be represented as an AI problem-solving process.

Initial State

The student starts the conversation by asking a question.

Goal State

The chatbot provides the correct answer and successfully satisfies the user's request.

Possible Actions

- Detect the user's intent.
- Extract entities from the query.
- Search the knowledge base.
- Generate the most appropriate response.

State Transition

The chatbot moves from one conversation state to another until the user's goal is achieved.

Conversation Flow

Start

↓

Greeting

↓

User Query

↓

Intent Detection

↓

Entity Extraction

↓

Knowledge Base Search

↓

Generate Response

↓

User Satisfied?

↓

Yes → End

No → Ask Again

Application of AI Search Technique

To improve dialogue navigation, the chatbot uses Artificial Intelligence search techniques.

Selected Search Strategy

Breadth First Search (BFS)

Why BFS?

Breadth First Search explores all possible conversation paths level by level before selecting the best response. It helps the chatbot identify the most suitable intent efficiently.

Advantages

- Finds the shortest conversation path.
- Improves response accuracy.
- Reduces incorrect intent selection.
- Easy to implement.
- Suitable for educational chatbot applications.

AI Reasoning

The chatbot compares the user's input with trained intents. After identifying the closest matching intent, it retrieves the relevant information and generates the response. If no suitable intent is found, the chatbot activates the Default Fallback Intent and asks the user to rephrase the question.

Testing and Evaluation

The chatbot was tested using different types of student queries.

Test Cases

User Query	Expected Result
Show my timetable	Displays timetable
Check my attendance	Shows attendance percentage
When are exams?	Displays examination schedule
Register for AI course	Shows registration details
Hostel information	Displays hostel details
Library timings	Shows library timings

Evaluation

The chatbot successfully answered most student queries with high accuracy. It responded quickly and correctly identified user intents. Unknown queries were handled through the Default Fallback Intent by asking users to provide more specific information.

Improvements

- Increase the number of training phrases.
- Add support for multiple languages.
- Integrate with the university database.
- Enable voice interaction.
- Improve response personalization using student login credentials.

Conclusion

The AI-Based Student Support Assistant developed using Dialog flow provides an efficient solution for handling common student queries. It reduces the workload of university staff by automating repetitive tasks and offers students quick, accurate, and reliable responses at any time. The chatbot demonstrates the practical application of Artificial Intelligence, Natural Language Processing, and search algorithms in solving real-world educational problems.

In the future, the system can be enhanced by integrating real-time university databases, voice assistants, multilingual support, and personalized student services. These improvements will make the chatbot more intelligent, user-friendly, and capable of providing a better student experience.